FIITJEE

ALL INDIA TEST SERIES

FULL TEST – V

JEE (Main)-2019

Time Allotted: 3 Hours **Maximum Marks: 360**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only one section: **Section-A**.

  Section-A (01 – 30, 31 – 60, 61 – 90) *contains 90 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Physics

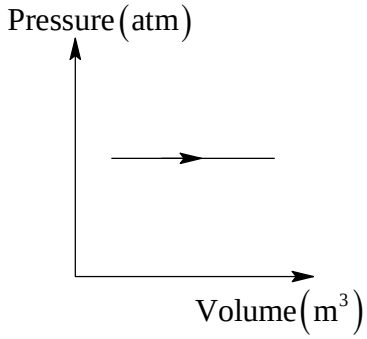
PART – I

SECTION – A
(One Options Correct Type)

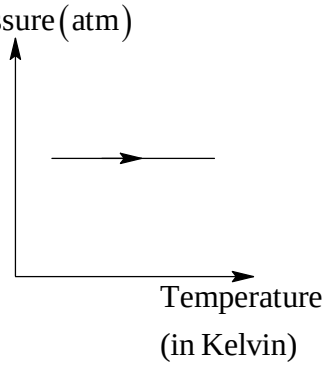
This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. All the graph shown are intended to represent the same thermodynamic process. One of them does it incorrectly. Pick it up:

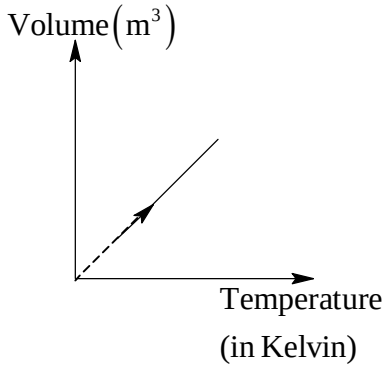
(A)



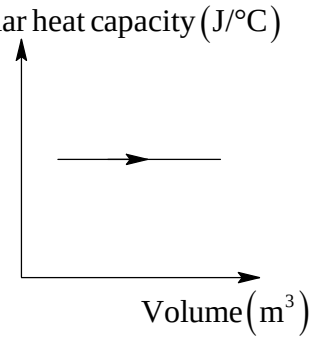
(B)



(C)



(D)



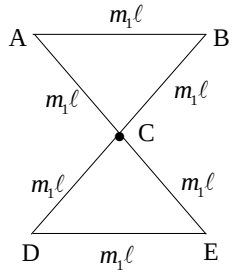
2. A string of length 1 meter, mass per unit length 0.01 kg/m and under tension of 100 newton is oscillating in first overtone. The frequency of oscillation is:

(A) 20 Hz (B) 50 Hz

(C) 100 Hz (D) 200 Hz

Space for Rough work

3. An arrangement of rods each of mass m and length l are welded (whereever required) to form a shape as shown. The moment of inertia about an axis passing through point C and perpendicular to the plane of figure is:



(A) ml^2 (B) $\frac{3}{2}ml^2$

(C) $3ml^2$ (D) $\frac{9}{2}ml^2$

4. Two particle performs simple harmonic motion with time period of 6 second and 7 second respectively. At t = 0 both were in their mean position and moving in the same direction. After what minimum non zero time they will be in phase again.
(A) 21 second (B) 42 second
(C) 63 second (D) 84 second

5. Two balls are projected with speed 10 m/s at an angle of 37° with the horizontal from the surface of two planets having ratio of mass $2:1$ and ratio of radii $1:2$. The ratio of time of flight is:
(A) $4:1$ (B) $1:1$
(C) $2:1$ (D) $1:8$

6. Three identical spheres A, B and C of radius r and each carrying charge q are placed on the vertices of an equilateral triangle of side $\ell\left(\ell >> r\right)$. The resultant force on B is F. Now an uncharged sphere is first touched with A and then with B and then removed. Now the resultant force on B is:

(A) F (B) $\frac{3\sqrt{7}}{8}F$

(C) $\frac{3}{4}\sqrt{7}F$ (D) $\frac{2\sqrt{7}}{4}F$

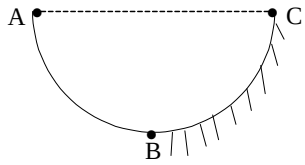
Space for Rough work

7. If in place of two slits in young double slit experiment four identical slits were used with equal spacing. Assuming coherent waves form these slits. The number of minima between two consecutive maximas will be:
(A) 1 (B) 2
(C) 3 (D) 4

8. In an Air conditioner compartment of a train if a material of thermal conductivity K and thickness 10 cm is required to fullfill the purpose. To do so there is a need of two glass each of thickness 1 cm with air gap of 8 cm between them. If thermal conductivity of glass and air are 1 W/mk and 0.01 W/mk then the value of K is:
(A) $0.0125 W/mk$ (B) $0.05 W/mk$
(C) $0.255 W/mk$ (D) $0.33 W/mk$

9. The end correction of a resonance column is 0.5 cm. If the shortest length resonating with the tuning fork is $\frac{21}{2}\text{cm}$, find the maximum lengths for resonance, if length of pipe is 100 cm.
(A) 59.1cm (B) 78.5cm
(C) 88.5cm (D) 98.5cm

10. A ball is released from the top of a semi cylindrical surface of which part AB is sufficiently rough and BC is smooth. Then:

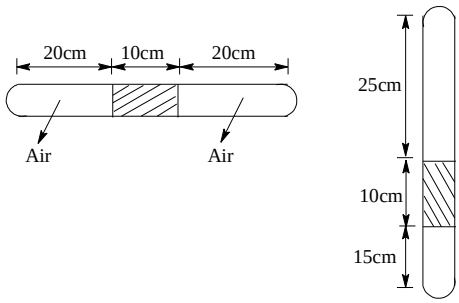


(A) The maximum angular velocity of the ball will be at point B only
(B) The maximum speed of centre of the ball will not be at point B
(C) Ball will never reach point C
(D) Ball may reach point C if friction is not sufficient to provide pure rolling.

11. A light ray falls perpendicular on one face of an equilateral prism, having refractive index $\sqrt{3}$. The angle of deviation is:
(A) $30°$ (B) $60°$
(C) $90°$ (D) $120°$

Space for Rough work

12. A horizontal tube contains mercury column of length 10 cm and rest is air on both sides of it. When the tube is vertical the length of air column changes as shown in the figure. The pressure when the tube is horizontal is (Assume temperature of air remains constant)



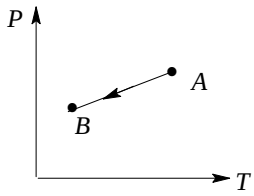
(A) $10^5\ \text{N/m}^2$

(B) $2.5\times10^3\ \text{N/m}^2$

(C) $0.255\times10^3\ \text{N/m}^2$

(D) $25.5\times10^3\ \text{N/m}^2$

13. An ideal gas undergoes a process $A \to B$ as shown on the pressure- temperature graph. The gas undergoes.



(A) Compression
(B) Expansion
(C) No change in volume
(D) Data insufficient

14. An object moves with velocity $\vec{v}_0 = 10\hat{i} + \hat{j} - 2\hat{k}$ in front of a plane mirror moving with velocity $\vec{V}_M = -5\hat{j}$. The velocity image is:

(A) $5\hat{i} + \hat{j} - 2\hat{k}$ (B) $10\hat{i} + \hat{j} + 2\hat{k}$

(C) $15\hat{i} + \hat{j} - 2\hat{k}$ (D) $10\hat{i} + \hat{j} - 2\hat{k}$

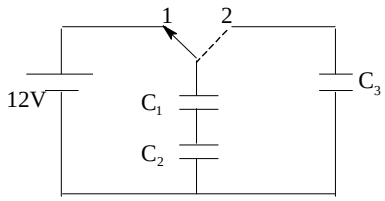
15. In the experiment of simple pendulum the percentage errors in length and time period in the measurement of acceleration due to gravity is 2% and 1% respectively. The maximum percentage error in acceleration due to gravity is:

(A) 1% (B) 2%

(C) 3% (D) 4%

Space for Rough work

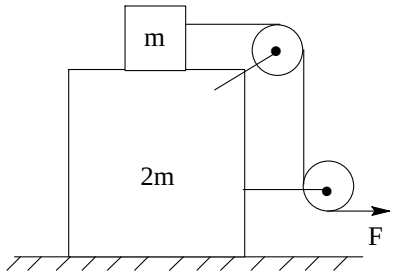
16. A capacitor $C_1 = 1\mu f$ and $C_2 = 2\mu f$ are charged with switch in position (1). Now switch is shifted to position (2). Capacitor $C_3 = 3\mu f$ is initially uncharged. The ratio of potential difference on C_1 and C_3 is:



(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) 1 (D) 2

17. A small block is placed on a bigger block as shown in the figure. All surfaces are frictionless and string and pulley are ideal. The acceleration of block of mass m is:



(A) F/2m (B) F/m

(C) F/3m (D) 2F/m

18. A current of 1 ampere is flowing on the sides of a square shaped wire frame of side $16\sqrt{2}cm$. The magnetic field at the centre of the square will be:

(A) $5\times10^{-6}Wb/m^2$ (B) $5\times10^{-5}Wb/m^2$

(C) $2.5\times10^{-5}Wb/m^2$ (D) $2.5\times10^{-6}Wb/m^2$

19. Hysteresis loop represent energy loss per cycle. Then energy loss is:

(A) More for steel than soft iron (B) More for soft iron than steel

(C) Equal for both (D) None of the above

Space for Rough work

20. STATEMENT-1: The temperature above which ferromagnetic material becomes paramagnetic is called curie temperature.
because
STATEMENT-2: Hysteresis is exhibited by ferromagnetic substance only.
(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

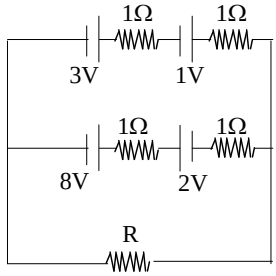
21. An astronomical telescope having focal length of objective 300 cm and of eyepiece 5 cm. It is focused to see an object 20 km from the objective. If final image is formed at infinity, then:
(A) Length of tube is 310 cm (B) Length of tube is 295 cm
(C) Angular magnification is 60 (D) Angular magnification is 100

22. A screw gauge with a pitch of 0.3 mm and a circular scale with 100 divisions is used to measure the thickness of a thin sheet of steel. What is the thickness of the sheet if main scale reading is 1mm and the 30th division coincides with the main scale line. (There is no zero error)?
(A) $1.9\,mm$ (B) $1.09\,mm$
(C) $1.1\,mm$ (D) $1.99\,mm$

23. Radiation of certain wavelength is incident on a metal plate of work function 5eV. The stopping potential is 2eV. Now if the wavelength is reduced 2 times, the new stopping potential will be:
(A) 6 Volt (B) 9 Volt
(C) 10 Volt (D) 12 Volt

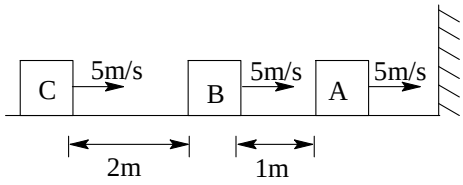
24. A close loop is placed in an uniform magnetic field. The current will not be induced in the loop if the loop:
(A) Rotate (B) Translate
(C) Remain stationary (D) Rolls

25. The circuit shown in the figure contains unidentical batteries and a bulb of unknown resistance R. For the bulb to have maximum brightness
(A) Its resistance should be 1Ω
(B) Its resistance should be 2Ω
(C) Current through bulb should be 6 Ampere
(D) Current through bulb should be 9 Ampere



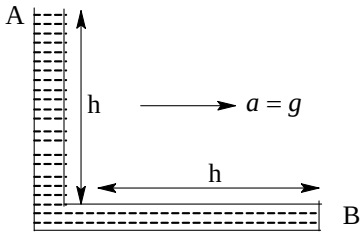
Space for Rough work

26. Three identical small blocks are moving towards a wall with spacing between them and velocities as shown in the figure. Assuming all collision to be elastic. The spacing between block B and C after all possible collisions is:



(A) 1m (B) 2m
(C) 3m (D) 4m

27. A L shaped tube containing liquid of density ρ is accelerated horizontally with acceleration as shown in the figure. The end of tube B is closed while end A is open. If the atmospheric pressure is P_0. Then the pressure at B is:



(A) $\frac{P_0}{2}$ (B) P_0
(C) $\frac{2}{3}P_0$ (D) $2P_0$

28. A TV tower has a height of 80 m. What is the maximum distance and area upto which this TV transmission can be received? Take radius of the earth as $6.4\times10^6 m$:
(A) $2816\,\text{km}^2$ (B) $2580\,\text{km}^2$
(C) $3215\,\text{km}^2$ (D) $3815\,\text{km}^2$

29. A transistor has a current amplification factor of 100. In a CE-amplifier circuit, the collector resistance is chosen as $10k\Omega$ and the input resistance is $1k\Omega$. The output voltage if input voltage is 0.01:
(A) 5 Volt (B) 10 Volt
(C) 15 Volt (D) 20 Volt

30. An elastic wire of length ℓ, area of cross-section A and Young's modulus Y having left end fixed and the other end is pulled by a constant force F, and the point of application of force is displaced by $\Delta\ell$. Then:
(A) Work done by force is $\frac{F\Delta\ell}{2}$
(B) Elastic potential energy stored in the wire $F\Delta\ell$
(C) Elastic potential energy stored in the wire $\frac{1}{3}F\Delta\ell$
(D) Work done by force is $F\Delta\ell$

Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Considering Ellingham diagram, which of the following metals can be used to reduce alumina?
(A) Mg (B) Zn
(C) Fe (D) Cu

32. Regarding cross-linked or network polymers, which of the following statements is ***incorrect***?
(A) Examples are Bakelite and melamine
(B) They are formed from bi-and tri-functional monomers.
(C) They contain covalent bonds between various linear polymer chains.
(D) They contain strong ionic bonds in their polymer chains.

33. Which is correct about amylose and amylopectin is
(A) Amylopectin have $1 \rightarrow 4\ \alpha -$ linkage and $1 \rightarrow 6\ \beta -$ linkage
(B) Amylose have $1 \rightarrow 4\ \alpha -$ linkage and $1 \rightarrow 6\ \beta -$ linkage
(C) Amylopectin have $1 \rightarrow 4\ \alpha -$ linkage and $1 \rightarrow 6\ \alpha -$ linkage
(D) Amylose is made up of glucose and galactose

34. A mixture of 2.3 g formic acid and 4.5 g oxalic acid (anhyd.) is treated with conc. H_2SO_4. The evolved gaseous mixture is passed through KOH pellets. Weight (in g)of the remaining gaseous product at STP will be
(A) $2 \cdot 8$ (B) $3 \cdot 0$
(C) $1 \cdot 4$ (D) $4 \cdot 4$

35. Which oxide of nitrogen is not a common pollutant introduced into the atmosphere both due to natural and human activity?
(A) N_2O (B) NO_2
(C) N_2O_5 (D) NO

Space for Rough work

36. The compound C_7H_8 undergoes the following reactions:

$$C_7H_8 \xrightarrow{3Cl_2/\Delta} A \xrightarrow{Br_2/Fe} B \xrightarrow{Zn/HCl} C$$

The product 'C' is

(A) 3-bromo-2, 4, 6-trichlorotoluene (B) o-bromotoluene
(C) m- bromotoluene (D) p- bromotoluene

37. Which one of the following ions exhibits d-d transition and paramagnetism as well?

(A) MnO_4^- (B) $Cr_2O_7^{2-}$
(C) CrO_4^{2-} (D) MnO_4^{2-}

38. For an ideal gas $\left(\frac{dU}{dT}\right)_p$ equals to:

(A) C_V (B) ΔS
(C) ΔG (D) ΔH

39. Iron exhibits BCC structure at room temperature. At $900°C$ and above, it transforms to FCC structure. The ratio of density of iron at room temperature to that at $900°C$ (assuming molar mass and atomic radii of iron remains constant with temperature) is.

(A) $\frac{3\sqrt{3}}{4\sqrt{2}}$ (B) $\frac{4\sqrt{3}}{3\sqrt{2}}$
(C) $\frac{\sqrt{3}}{\sqrt{2}}$ (D) $\frac{1}{2}$

40. For an isothermal process for an ideal gas, ΔU is equal to:

(A) Adiabatic work (B) Isothermal work
(C) Isochoric work (D) Isobaric work

41. For the Tyndal effect which is incorrect:

(A) The diameter of the dispersed particle is not much smaller than the wave length of light used.
(B) The refractive indices of the dispersed phase and dispersion medium differ in great magnitude
(C) Information about the size of colloidal particle can't be obtained by ultra microscope.
(D) None of the above

Space for Rough work

42. $4\,\mathrm{P}\ (\mathrm{s,white}) + 3O_2(g) \longrightarrow P_4O_6(s) \quad \Delta H^\circ = x$

$4\,\mathrm{P}\ (\mathrm{s,red}) + 3O_2(g) \longrightarrow P_4O_6(s) \quad \Delta H^\circ = y$

$\mathrm{P_4O_6(s) + 2O_2(g) \longrightarrow P_4O_{10}(s)} \quad \Delta \mathrm{H^\circ = z}$

The standard enthalpy of formation of $\mathrm{P_4O_{10(s)}}$ at 298 K will be:

(A) x+z (B) y+z

(C) x+y-z (D) x-y+z

43. Which of the following statements about LDP is false?
(A) Its synthesis requires high pressure
(B) It is poor conductor of electricity
(C) Its synthesis requires dioxygen or a peroxide initiator as catalyst
(D) Its used in the manufacture of buckets dustbin etc.

44. Distillation technique is most suitable for separating glycerol from spent lye in soap industry is:
(A) Simple distillation (B) Fractional distillation
(C) Steam distillation (D) Distillation under reduced pressure.

45. The phosphorous in +3 oxidation state is:
(A) Orthophosphorous acid and Pyrophosphorous acid
(B) Pyrophosphorous acid and Hypophosphorous acid
(C) Orthophosphorous acid and Hypophosphoric acid
(D) Pyrophosphorous acid and Pyrophosphoric acid

46. Dilute solution of KOH reacts with I_2 and forms:

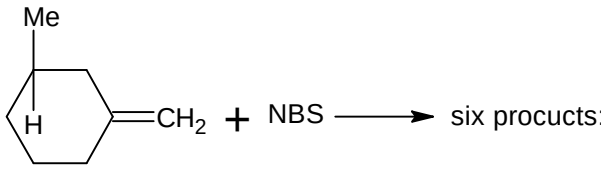
(I) KI (II) KI_3

(III) KIO

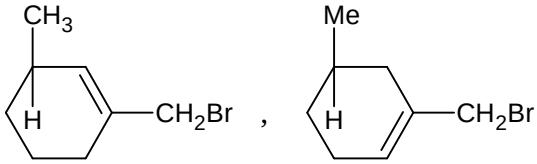
(A) II, II (B) I, III

(C) I (D) I, II, III

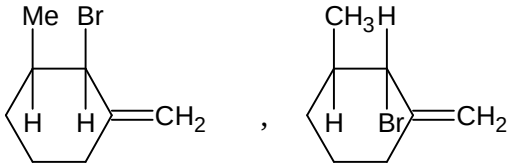
Space for Rough work

47. 

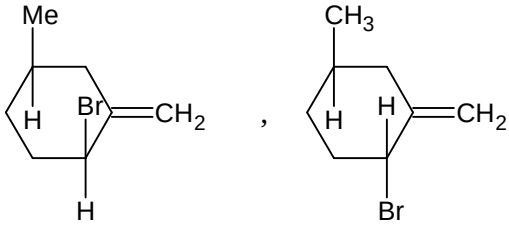
Which of the following statements is incorrect about products?

(A) 

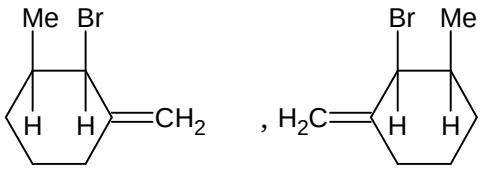
are two possible products.

(B) 

are other two possible products

(C) 

are other two possible products.

(D) 

are other two possible products

Space for Rough work

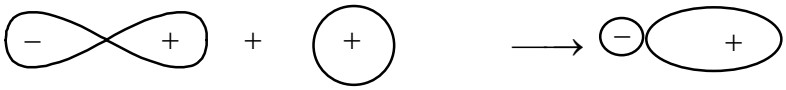
48. $0.32g$ of CaC_2 reacts with water produces gas X. The gas X combination with O_2 produce CO_2. The volume of CO_2 at STP:

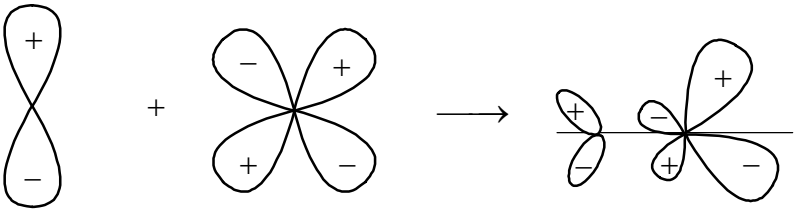
(A) $112\,mL$ (B) $484\,mL$

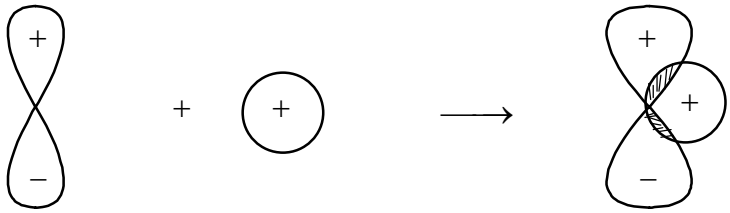
(C) $224\,mL$ (D) $448\,mL$

49. Which of the following combination of atomic orbitals result into non-bonding?

(A) 

(B) 

(C) 

(D) 

Space for Rough work

50. Consider the Vander Waal's equation of state $\left(p+\frac{a}{V^2}\right)(V-b)=RT$ correct statements is:

(A) Value of 'a' of N_2 is larger than that of NH_3 due to higher molecular weight of N_2

(B) Gas 'A' has Vander Wall's constants 'a' $\left(L^2\,atm/mol^2\right)$ and 'b'$\left(L/mol\right)$ as 7 and 4.98×10^{-2}, this gas cannot be liquid at temperature $200°C$ however high pressure may be applied.

(C) At high pressure above equation reduces to $Z=1-\frac{b}{RTV}$

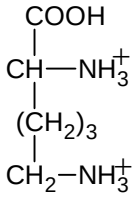
(D) If $Z<1$, attractive forces are dominating among the gaseous molecules

51. While testing BO_3^{3-}, there is green-edges flame on heating the salt with conc. H_2SO_4 and CH_3OH. Green colour is of

(A) $\left(CH_3\right)_3B$ (B) $\left(CH_3O\right)_3B$

(C) B_2O_3 (D) H_3BO_3

52.

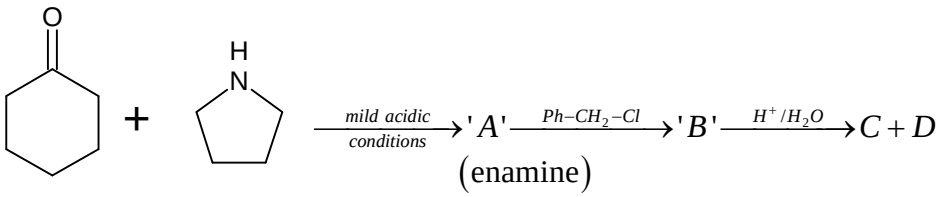


Lysine has $pK_{a,1}, pK_{a,2}, pK_{a,3}$ values 2, 9 and 11 respectively. Then value of its isoelectric point $\left(pH_{\text{ isoelectric}}\right)$ is:

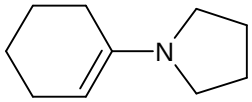
(A) 7.22 (B) 5.565

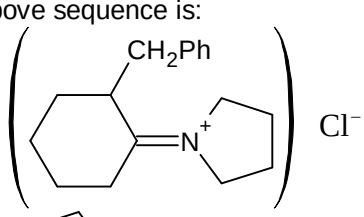
(C) 6.355 (D) 10

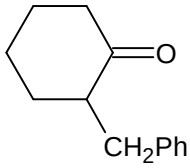
Space for Rough work

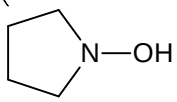
53. 

The incorrect statement about various organic products in above sequence is:

(A) A is 

(B) B is 

(C) C is 

(D) D is 

54. An orange coloured compound (A) when treated with $NaCl$ & conc. H_2SO_4 gives red brown vapour (B). This vapour when passed through $NaOH$ gives yellow coloured solution (C). On addition of CH_3COOH and $(CH_3COO)_2 Pb$ to this yellow solution, yellow colored ppt (D) is obtained. Which of the following compound is not resemble with above information:

(A) 'A' is $K_2Cr_2O_7$ (B) 'B' is CrO_2Cl_2

(C) 'C' is Na_2CrO_4 (D) 'D' is $PbCl_2$

55. A current of $0.55\,Amp$ deposits $0.55\,gm$ of a metal in 100 minutes. Calculate atomic mass of the metal if equivalent weight $= \frac{M.W.}{2}$?

(A) 100 (B) 45

(C) 32.35 (D) 144.75

Space for Rough work

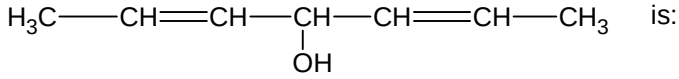
56. Consider following transformation

$\left[\mathrm{Co}\left(\mathrm{H_2O}\right)_{\mathrm{x}}\right]^{2+}$ (Pink complex) $\xrightarrow[\mathrm{HCl}]{\text{Excess}}\left[\mathrm{CoCl_y}\right]^{n-}$ (Blue complex)

Find the value of expression $\frac{x+y}{n}$.

(A) 5 (B) 8
(C) 10 (D) 7

57. Total number of stereoisomers of the compounds



is:

(A) 2 (B) 3
(C) 4 (D) 8

58. $\mathrm{AgNO_3} \xrightarrow{\Delta} (\mathrm{W}) + (\mathrm{X}) + O_2$

$(\mathrm{X}) + \mathrm{H_2O} \longrightarrow \mathrm{HNO_2} + \mathrm{HNO_3}$

$(\mathrm{W}) + \mathrm{HNO_3} \longrightarrow \mathrm{Y} + \mathrm{NO} + \mathrm{H_2O}$

$(\mathrm{Y}) + \mathrm{Na_2S_2O_3}\,(excess) \longrightarrow (Z) + \mathrm{NaNO_3}$

Identify (W) to (Z).

(A) $W = Ag \quad X = N_2O \quad Y = AgNO_3 \quad Z = Na_2\left[\mathrm{Ag}\left(\mathrm{S_2O_3}\right)_2\right]$

(B) $W = AgO_2 \quad X = NO \quad Y = AgNO_3 \quad Z = Na_2\left[\mathrm{Ag}\left(\mathrm{S_2O_3}\right)_2\right]$

(C) $W = Ag \quad X = NO_2 \quad Y = AgNO_3 \quad Z = Na_3\left[\mathrm{Ag}\left(\mathrm{S_2O_3}\right)_2\right]$

(D) $W = Ag_2O \quad X = N_2 \quad Y = AgNO_3 \quad Z = Na_2\left[\mathrm{Ag}\left(\mathrm{S_2O_3}\right)_2\right]$

Space for Rough work

59. Which of the following statements is/are correct?
(i) Sulphur solution is an example of macro molecular colloid.
(ii) Adsorbent used in gas masks is activated charcoal
(iii) AgI is used for producing artificial rain by seeding because AgI has crystal structure similar to ice.
(iv) Alums purify muddy water by coagulation of colloidal sized clay & sand.
(v) Physisorption occurs because of van der Waals forces.
(vi) More easily liquefiable gases are adsorbed readily. (For physisorption)
(vii) Under high pressure it results into multimolecular layer on adsorbent surface. (For physisorption)
(viii) Enthalpy of physisorption adsorption ($\Delta H_{adsorption}$) is low and positive.

(A) 6 (B) 7
(D) 8 (D) 5

60. When a liquid that is immiscible with water was steam distilled at 95.2°C at a total pressure of 748 torr, the distillate contained 1.25 g of the liquid per gram of water. The vapour pressure of water is 648 torr at 95.2°C, what is the molar mass of liquid?

(A) $8\,g/mol$ (B) $166\,g/mol$
(C) $146\,g/mol$ (D) None of these

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If $f(x) = \sin^4 x + \cos^4 x - \frac{1}{2}\sin 2x$, then the range of f(x) is

(A) $\left[0, \frac{3}{2}\right]$ (B) $\left[-\frac{1}{2}, \frac{7}{2}\right]$

(C) $\left[0, \frac{9}{8}\right]$ (D) $\left[\frac{3}{4}, \frac{7}{8}\right]$

62. The value of $\prod_{r=1}^{7} \cos\frac{\pi r}{15}$ is

(A) $\frac{1}{64}$ (B) $\frac{1}{128}$

(C) $\frac{1}{32}$ (D) $\frac{1}{4}$

63. The exhaustive intervals of real values of x such that $\sqrt{12-4x} > 1+\sqrt{4x+4}$ is

(A) $\left[1-\frac{\sqrt{31}}{8}, 1+\frac{\sqrt{31}}{8}\right)$ (B) $\left[-1, 1+\frac{\sqrt{31}}{8}\right)$

(C) $[-1, 3]$ (D) $\left[-1, 1-\frac{\sqrt{31}}{8}\right)$

64. If $P = \int_{-1}^{2}\left(\frac{\sqrt{x^2+1}+x-1}{\sqrt{x^2+1}+x+1}\right)dx$ and $Q = \int_{1}^{2}\left(\frac{x}{\sqrt{x^2+1}+1}\right)dx$ then $\frac{P}{Q}$ is equal to

(A) $\ln\frac{2}{5}$ (B) $\frac{2}{3}$

(C) $\frac{1}{2}$ (D) 1

Space for Rough work

65. $\int \frac{e^x(x-2)}{x\left(x^2+e^x\right)}$ $\forall$ x > 0 is equal to

(A) $\ln\left(1+\frac{e^x}{x^2}\right)+c$ (B) $\ln\left(-\frac{1}{2}+\frac{e^x}{x^2}\right)+c$

(C) $\ln\left(2+\frac{e^x}{x^2}\right)+c$ (D) $\ln\left(x+\frac{e^x}{x^2}\right)+c$

66. The number of positive 6 digit integers such that the sum of their digits is 9 and four of its digits are 1, 0, 0, 5 is

(A) 60 (B) 120

(C) 180 (D) 210

67. Consider the parabola $x^2 + 4x - 2y + 6 = 0$. If its chord AB $2mx - 2y + (4m + 3) = 0$ intersect its axis at K and directrix at M, then AM, KM and BM are in

(A) A.P (B) G.P

(C) H.P (D) none of these

68. $\lim_{x\to\infty}\left(\frac{x^{2013}}{e^{3x}}+\left(\cos\frac{2}{x}\right)^{x^2}\right)$ is

(A) 0 (B) e^3

(C) e^2 (D) e^{-2}

69. If a, b, c denotes the sides of a ΔABC and $a + b - c = 2$ and $2ab - c^2 = 4$, then

(A) $\sin A + \sin B + \sin C < \frac{3\sqrt{3}}{2}$ (B) $\cos A + \cos B + \cos C = 1$

(C) $\frac{a}{b}+\frac{b}{c}+\frac{c}{a} > 3\sqrt{3}$ (D) $a^2 + b^2 + c^2 = 12$

70. If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$, then the value of $\frac{xy\sqrt{1-z^2}+yz\sqrt{1-x^2}+zx\sqrt{1-y^2}}{\sqrt{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}}$ is equal to

(A) 0 (B) 1

(C) 2 (D) 3

Space for Rough work

71. If $\alpha, \beta(\alpha < \beta)$ are the roots of the equation $g(x) = ax^2 + bx + c = 0$ and f(x) be an even function, then $\int_{\alpha}^{\beta} \frac{e^{f\left(\frac{g(x)}{x-\alpha}\right)}}{e^{f\left(\frac{g(x)}{x-\alpha}\right)} + e^{f\left(\frac{g(x)}{x-\beta}\right)}} dx$ is equal to

(A) $\frac{1}{2}(\beta - \alpha)$ (B) $\frac{1}{2}(\beta^2 - \alpha^2)$

(C) $\frac{1}{2}(\beta + \alpha)$ (D) 0

72. The equation $x^2 + (1 - 2^{2013})x + 2^{2012}(2^{2012} - 1) - 2 = 0$ have roots α and β then the value of $(\alpha - \beta)^{2014}$ is

(A) $(2^{2012} - 1)1007$ (B) $(3)^{4018}$

(C) 3^{2014} (D) $(2)^{1007}$

73. The value of ${}^nC_1 + 3^2\ {}^nC_3 + 5^2\ {}^nC_5 + \ldots\ldots$ is

(A) $(n - 1)2^{n-1}$ (B) $n(n + 1)2^{n-3}$

(C) $n(n + 2)2^{n-2}$ (D) $n(n + 1)(n + 2)2^{n-4}$

74. If $(1 + 3 + 5 + \ldots.. + p) + (1 + 3 + 5 + \ldots. + q) = (1 + 3 + 5 + \ldots.. r)$ where each set contains the sum of consecutive odd integers then the smallest possible value of p + q + r is

(A) 23 (B) 21

(C) 29 (D) 31

75. Let $A = \{(x\ y) : x + y = p\}$, $B = \{(x, y) : x^3 + y^3 < p\}$ be two sets such that $A \cap B = \phi$ for any real number p, then the exhaustive intervals of p is

(A) $[-2, 2]$ (B) $\{0\} \cup [2, \infty)$

(C) $(-\infty, -2] \cup \{0\}$ (D) $\{x!\ x \in N \wedge x \leq 3\}$

76. Let C be incircle of ΔABC, if tangents are drawn parallel to the sides of the Δ such that t_1, t_2, t_3 be the lengths of tangents which lie inside the given triangle parallel to sides a, b, c respectively the $\frac{t_1}{a} + \frac{t_2}{b} + \frac{t_3}{c}$ is equal to

(A) 0 (B) 1

(C) 2 (D) 3

Space for Rough work

77. The value of ${}^nC_0 + {}^nC_1 + 5\,{}^nC_2 + 11\,{}^nC_3 + 17\,{}^nC_4$.....$(n+1)$ terms

(A) $(3n+5)\cdot 2^n + n + 7$ (B) $(3n-7)\cdot 2^n + 2n + 8$

(C) $(2n+3)\cdot 2^n - 11$ (D) $\left(2^n+3\right)\cdot 5^n + 7$

78. Let x + y - z – 4 = 0 be the equation of plane and A be a point (1, 2, –3). If line L passes through (1, 2, 3) with direction ratios 3 : –1 : 4. If distance of A from line measured parallel to the plane is d_1 and if distance of the plane measured parallel to line is d_2 then $\left(\sqrt{d_1^2 - d_2^2}\right)$ is

(A) 1 (B) 5

(C) 3 (D) 10

79. If $a_1x^3 + b_1x^2 + c_1x + d_1 = 0$ and $a_2x^3 + b_2x^2 + c_2x + d_2 = 0$ have a pair of common positive repeated roots. If the value of $\begin{vmatrix} 3a_1 & 2b_1 & c_1 \\ 3a_2 & 2b_2 & c_2 \\ a_2b_1 - a_1b_2 & c_1a_2 - c_2a_1 & d_1a_2 - a_2d_2 \end{vmatrix}$ is

(A) 3 (B) 2

(C) 1 (D) 0

80. A random variable X has the following probability distribution values of X

X:	0	1	2	3	4	5	6	7
P(X):	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2 + k$

then which is true

(A) $\dfrac{k+1}{P(x<6)} = \dfrac{10}{81}$ (B) $\dfrac{P(0<x<5)}{P(x\geq 6)} = \dfrac{81}{19}$

(C) $\dfrac{k-1}{P(x\geq 6)} = \dfrac{2}{23}$ (D) $\dfrac{P(0<x<4)}{P(x<6)} = \dfrac{4}{493}$

81. If $f(x) = \int_0^x \left(e^t + 3\right)dt$, then which of the following is always true $\forall\ x \in R$

(A) $f(x-3) - f(x-1) \leq 11$ (B) $f(x+1) - f(x-1) < 4$

(C) $f(x+3) - f(x-5) > 24$ (D) $f(x) + f(x+1) < -21$

Space for Rough work

82. Consider $f : R^+ \to R$ such that $f(3) = 1$ for $a \in R^+$ and $f(x).f(y) + f\left(\frac{3}{x}\right)f\left(\frac{3}{y}\right) = 2f(xy)\ \forall\ x, y \in R^+$, then f(97) can be

(A) 1 (B) –1

(C) 2 (D) 97

83. If $\alpha, \beta, \gamma, \delta$ be first 4 positive solutions of $\sin x = \frac{1}{4}$ with $\alpha < \beta < \gamma < \delta$, then value of $\sin\frac{\delta}{2} + 2\sin\frac{\gamma}{2} + 3\sin\frac{\beta}{2} + 4\sin\frac{\alpha}{2}$ is

(A) $\sqrt{5}$ (B) $2\sqrt{5}$

(C) $\frac{3}{4}$ (D) $\frac{5}{4}$

84. If e_1 and e_2 are the roots of the equation $x^2 - ax + 2 = 0$ where e_1, e_2 are the eccentricities of an ellipse and hyperbola respectively then the value of a belongs to

(A) $(3, \infty)$ (B) $(2, \infty)$

(C) $(1, \infty)$ (D) $(-\infty, 1) \cup (1, 2)$

85. If domain of f(x) is [1, 3], then find the domain of $f(\log_2 (x^2 + 3x - 2))$

(A) $[-5, -4] \cup [1, 2]$ (B) $[-13, -2] \cup \left[\frac{3}{2}, 5\right]$

(C) $[-4, 1] \cup [2, 7]$ (D) $[-3, 2]$

86. ΔABC is inscribed in a unit circle such that three internal angle bisectors of ΔABC are extended to intersect circumcircle at P, Q, R respectively then maximum value of $AP\cos\frac{A}{2} + BQ\frac{B}{2} + CR\cos\frac{C}{2}$ is equal to

(A) $\sqrt{3}$ (B) $3\sqrt{3}$

(C) $\frac{\sqrt{3}}{2}$ (D) 1

Space for Rough work

87. If $f : R \to R$ such that $f(x) = \frac{5}{lx^3 + x^2 + x + m}$ then (l, m) can be

(A) (1, 2) (B) (0, 1)

(C) (0, –3) (D) $\left(-3, \frac{3}{2}\right)$

88. For any complex number z in the argand plane, the minimum value of $|z| + |z - e^{i\alpha}| + |z - 5e^{i\alpha}| + |z - 7e^{i\alpha}| + |z - 32e^{i\alpha}|$ is

(A) 22 (B) 32

(C) 38 (D) 42

89. Let A = {1, 2, 3}, then the relation R = {(1, 1), (1, 2), (2, 1)} on A is

(A) reflexive (B) transitive

(C) symmetric (D) none of these

90. A flag staff of 5 m height stands on a building 25 m high. At an observer at a height of 30 m, the flag staff and the building subtend equal angles. The distance of the observer from the top of flag staff is

(A) $\frac{5\sqrt{3}}{2}$ (B) $5\sqrt{\frac{3}{2}}$

(C) $\frac{2}{3}$ (D) $\frac{5}{3}$

Space for Rough work